

Jitin Nair

AI SYSTEMS ARCHITECT · MULTI-AGENT ORCHESTRATION SPECIALIST

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PROFESSIONAL SUMMARY

AI Systems Architect with **200+ autonomous agents in production** and a decade of engineering context spanning learning science, data systems, and agentic AI. Specialised in **multi-agent orchestration** (LangGraph, Model Context Protocol), **RAG architecture**, and institutional-grade autonomous platforms. Built Predator Nexus (HFT engine, 91.2% regime accuracy), Reality-Check (governance for 147 agents, 96% fleet coverage), LocalMind (local RAG, 98% factual grounding), and Smartslate (AI-native L&D platform, 45-second blueprint generation). A background in instructional design and human performance provides rare depth in **human-AI collaboration systems** and explainability requirements.

PROFESSIONAL EXPERIENCE

AI Systems Architect

Mar 2025 – Present

Independent Consultant · Bangalore, India

- Deployed **200+ autonomous AI agents** across production multi-agent orchestration platforms, engineered on LangGraph with stateful DAG execution and real-time Redis state pulses.
- Architected **Model Context Protocol (MCP)** integrations enabling cross-agent tool sharing across **147 agent instances** — achieving 96% fleet coverage for truth-verification protocols with zero active violations post-deployment.
- Built **Predator Nexus V4.0**: institutional-grade HFT engine fusing Multivariate Gaussian HMM, LangGraph DAG orchestration, and Bayesian posterior inference at **8.4ms p99 latency** — 91.2% regime accuracy, 70.2% validated strategy yield across 10,000+ cycles.
- Engineered **LocalMind**: privacy-first local RAG platform indexing **1,000 files per minute** via SHA-256 delta indexing, achieving 98% factual grounding and 1s p99 hybrid-search latency.
- Designed **Smartslate**, an AI-native L&D platform running **500+ daily automated jobs** that compresses full programme discovery from weeks to **45 seconds** — multi-provider AI (Claude, ChatGPT, Perplexity) with human-in-the-loop validation at every stage.
- Applied **chain-of-verification (CoVe)** and factual-grounding protocols across agent fleets to eliminate hallucination in downstream decisions — reducing fictional data generation to below 2.2% across monitored deployments.

Instructional Designer | Lead: Video Based Learning

Sep 2022 – Mar 2025

Moody's Ratings · Bangalore, India (Global Remit)

- Built **automated content production pipelines** for a global credit ratings agency — reducing build time by **60%** and achieving **90%+** module completion at scale.
- Developed template standardisation systems and workflow automation, establishing **repeatable production infrastructure** that served as the foundation for later AI-pipeline thinking in Smartslate.
- Applied **learning systems thinking** to complex financial knowledge architecture — directly informing subsequent AI agent design patterns for knowledge retrieval and structured output.

Instructor Analyst → Senior Analyst

Jan 2019 – Sep 2022

Accenture · Bangalore, India

- Engineered a **1,400-line VBA automation suite** for pan-India training-needs audits — generating pivot tables, charts, and automated feedback emails, eliminating weeks of manual overhead across the network.
- Built multiple **hybrid learning systems** combining modality-matching with measurable outcome tracking — reducing F&A client learning time by **70%** while maintaining 54% knowledge retention.
- Led L&D for **300+ employees across 16 client teams**, delivering \$140K+ in training cost savings through systematic automation and blended design.

Senior Executive — Operations & Training

May 2015 – Dec 2017

[24]7.ai · Bangalore, India

- Managed high-volume back-end operations for a US retail and e-commerce client, running training, QA, and refresher programmes maintaining team performance above benchmark.
- Recognised as **top performer three consecutive months**; identified for leadership track within 24 months — earliest accelerated path available.

SYSTEMS ARCHITECTURE PORTFOLIO

Predator Nexus V4.0

Lead Architect · Institutional-Grade HFT Engine

Bayesian multi-agent orchestration ecosystem for autonomous XAU/USD trading — 7 specialised agents (Hermes, Argus, Athena, Apollo) coordinated via LangGraph DAG with sub-10ms posterior inference.

- 8.4ms p99** end-to-end signal-to-socket latency · **91.2%** Multivariate HMM regime accuracy
- 70.2%** validated strategy yield across 10,000+ cycles · 5,000+ msg/sec ProtoBuf throughput, zero packet loss

Python 3.13

LangGraph

TimescaleDB

Bayesian HMM

cTrader OpenAPI

Redis Streams

Numba

Reality-Check Governance Protocol

Lead Architect · Distributed AI Governance

Governance enforcement layer for 147 AI agents — implements truth-only protocols, automated policy injection, and audit logging to eliminate hallucination and fictional data generation at fleet scale.

- 96% fleet coverage** (141/147 agents) · **3-minute** full-fleet policy rollout · zero active violations post-deployment
- Automated policy injection + audit trail with full observability across the distributed agent network

Python

Node.js

Bash

YAML

Audit Logging

MCP

LocalMind

Lead Architect · Privacy-First Local RAG

Local-first RAG intelligence platform — SHA-256 delta indexing, hybrid semantic search, and source-only grounding for 100% on-premise operation with no data leaving the machine.

- 1,000 files/min** delta-indexing · **1s p99** hybrid-search latency · **98%** factual grounding (source-only)
- Fully air-gapped operation — all inference, retrieval, and storage runs on-device

Next.js

Python

pgvector

Ollama

SHA-256 Delta Index

Supabase

Smartslate · AI-Native L&D Platform

Lead Architect & L&D Designer · smartslate.io

AI-first platform automating the full L&D lifecycle — 7-stage contextual discovery pipeline with multi-provider AI and human-in-the-loop validation at every stage boundary.

- 45-second** tailored learning blueprint generation (vs. 2–4 week traditional sprints)
- 500+ daily automated jobs** with zero manual intervention · Kirkpatrick Level 4 measurement standard

Next.js 15

Supabase

Claude

ChatGPT

Perplexity

PostgreSQL

TECHNICAL STACK

AI / AGENTIC	LangGraph · Model Context Protocol (MCP) · Claude · ChatGPT · Gemini · RAG Architecture · Vector DBs · Prompt Engineering · Chain-of-Verification
LANGUAGES	Python 3.13 · TypeScript · Node.js · Bash
DATA & SYSTEMS	TimescaleDB · PostgreSQL · Redis Streams · pgvector · Prometheus · Docker · GitHub Actions
FULL-STACK	Next.js 15 · React · FastAPI · Supabase · Playwright · Tailwind CSS

EDUCATION & LANGUAGES

Bachelor of Commerce (B.Com)

Sindhi College of Commerce · Bangalore, India

2012 – 2015 · Self-taught programmer; built first web applications during undergraduate years

English — Native Malayalam — Native Hindi —
Fluent Tamil — Conversational Kannada —
Conversational